Chapter 8
Student's Electronic Card: A Secure Internet Database System for University Management Support

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Abstract. In the paper, design concept, architecture and functionality of the "Students' Electronic Card" computer system are presented. The system was developed in 1998-2007 and deployed at the Faculty of Electrical, Electronic, Computer and Control Engineering, Technical University of Lodz, Poland. It aids to efficiently organize activities of the community, including 4500 students along with 450 faculty and staff members. The core elements of the system are internet database (Oracle, Postgress) and a network of different functionality terminals (class assignment, cost analysis, marking, student portal, etc.). These terminals are used to enter, edit or retrieve data related to the management of teaching, research and other activities at the faculty, including funds allocation and cost analysis. A number of users of different roles and privileges are defined in the system. Smart cards are issued to users to store their personal secret keys for data encryption and electronic signature generation while communicating with the database. The system has been continuously updated and expanded for more than 10 years. One of the recent projects in this area is "Electronic Education Connectivity Solution", aimed at providing support to Bologna processed-governed student mobility between European universities, funded from 7FP UE program.

8.1 Introduction

While the clear mission of a university is education, it is certainly a very large business enterprise, entailing huge administrative activity [1]. From a managerial point of view, the university campus is a rather large and complex business organization. It has to provide many diverse services to a large number of people who are engaged

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in a huge variety of activities [2]. The services have to operate smooth to the satisfaction of their users; they should help, not disrupt, the statutory user activities and they should be cost-effective [1].

The core of campus services is now a database system or a network of local databases, seamlessly connected to each other. Most of the information stored includes user individual data, such as personal information or marks acquired by students. Such sensitive individual information should not be disclosed to unauthorized persons. Thus users transfer the information to the database and read it from the database through secure communication channels. To make the channel secure, the user must be identified by a unique ID code, used e.g. to encrypt the information and/or attach electronic signature to the data chunks [3], [4].

Many campus services include payments, e.g. for parking, printing, or operating vending machines. Cashless purchase is definitely preferred to cash payments, as it reduces the costs of handling and moving cash. Smart card used as an electronic purse [5], [6], [7] provides a means of eliminating cash flow from campus transactions. Yet another vital issue relates to access control - to rooms, facilities, libraries, computer networks. To be authorized for a service, the user has to prove his/her identity or present a certified permit which can be verified in the system by reading information stored in a campus card memory.

New opportunities and challenges are generated by the process of integration of European educational systems [8]. The idea of student and staff mobility between universities has gained a wide acceptance as expressed by the Bologna Declaration [9]. To implement this idea in practice, one needs a common ID number that could be stored in a campus card memory, along with the study-relevant personal information, to be carried by a student to any university he/she is actually enrolled to.

All the examples mentioned above indicate the need for a secure information storage that can be carried by its user/owner everywhere. Over the years, a plastic card has become such a storage device. Depending on application, a magnetic stripe or an electronic memory/microprocessor chip is embedded in the card. The cards equipped with a microprocessor, and an operating system, are called "smart cards". Unlike magnetic stripe cards, smart cards carry all necessary information and processing functions on the card. Therefore, they do not require access to remote databases at the time of the transaction [10]. The cost of a single, non-personalized card may range from 0.2 eurocents for a magnetic stripe, through 2.50 euro for a memory card to 10 euro for a smart card. There were over 900 million credit cards (of magnetic stripe type mainly) in circulation in 2006, global production of smart cards surpassed 5 billion in 2008 [11]. Major uses include providing enhanced financial services, increasing the security and flexibility of cellular phones.

It is evident that plastic cards have far-reaching impact on e.g. financial services, transportation, telecommunications, and health. A smart card is now in fact a wearable computer that serves as a secure storage of personal information in almost every place and time of our activity.

The smart card is of course not a panacea for all user authentication problems. Since the card can be stolen together with a PIN code to it, then in high security applications there is a need for other, supplementary techniques of people identification.

Most of these techniques are based on measuring personal characteristics of user body or behavior, such as fingerprint, retina blood vessel patterns or voice parameters. These types of biometric information can be used to enhance (augment) or replace information stored on the card. Development of the biometric authentication/identification instruments integrated with campus card systems requires new sensors for reliable measurement of personal information and new methods of signal processing, image analysis and pattern recognition. These issues again fall within the range of interest of the Students' Electronic Card system designers and developers [12].

This paper provides information about architecture, functions and implementation issues of the Students' Electronic Card system. The system is an extensively verified, working example a vital, widespread and still growing application of secure internet database systems designed for the support of management of a very complex enterprise - a university.

8.2 Project Background

The idea of applying electronic cards to store student marks and help in the management of teaching at the Faculty of Electrical, Electronic, Computer and Control Engineering, Technical University of Lodz (EECCE TUL) was first considered in 1996-98. The need of introducing electronic documentation was well recognized at the Faculty at that time. The four-fold increase of the number of students in a few years (to about 4500 students, which made the EECCE the largest TUL faculty), caused many organizational problems to academic and administrative staff. These problems endangered quality of student services and needed quick solution. Initially, memory cards were considered [1] to serve as a form of electronic student book of records. Each student was expected to be issued a plastic card with her/his photo for visual identification, and a digital memory of capacity sufficient to store all data during the whole study. This card was to be carried in a pocket of the paper book of records (which can not be eliminated due to national regulations). Teachers were to write in the marks both to the paper book and to the card memory, the latter with the use of a card reader connected to a computer.

Prototype card readers and software modules were designed, developed and used for tests which confirmed feasibility of the system. The design was based on Siemens cards [13]. After assessment of the investment, material and exploitation costs [1] of the enterprise, accounting for the size of the Faculty (4500 students and more than 450 faculty and administrative/technical staff) the basic assumptions concerning the system specifications were modified. Giving each student a card for marks duplication and further off-line storage in the system was considered too costly. A decision was made the marks will be stored in a common internet database accessed by teachers and administrative staff. The microprocessor cards would be issued to teachers and administrative staff only, in the first phase of system deployment at the Faculty. Main function of the card was the storage of user identification data and the

private keys used for encryption and digital signature of the data exchanged between the internet database and the system terminals. Based on this decision, a comprehensive faculty information processing system has been developed and expanded, now involving the students also as passive users (i.e. with no authorization to write the crucial data into the base, yet). It is expected that wide introduction of electronic student ID card [14] will allow granting the students the right to write the data, e.g. to electronically various forms, make payments, etc.

In general, language and legal issues have severe impact on university management system design. For this reason, there are no universal, comprehensive, global solutions available. The systems are developed locally, and then adapted to the needs of information exchange with collaborating foreign universities. At the time the development of the SEC system started, the only solution available on the Polish market was HMS product by Kalasoft company [15]. The HMS allowed for collecting of student marks and aided the timetable planning. However, it did not support the ECTS framework. HMS was not an internet solution; it operated on local area network only. Currently, there is a number of software solutions which support different fields of activity of higher education institutions. Big companies such as SAP, Oracle and Microsoft offer their well-known ERP systems, more or less customized to the higher education requirements. Smaller software developers propose dedicated, integrated solutions, e.g. Sygnity Edukacja.CL [16], Kalasoft HMS Solution [15], Comarch CDN Egeria Edukacja [17], Partners in Progress Uczelnia.XP [18]. Some systems e.g. SUSZI [19] or USOS [20], have been developed by universities, or by university consortia. Comparative in-depth analysis concerning functionality and security of these systems has not been performed and published yet, to the authors knowledge. From the point of view of prospective system users, such an analysis is much needed. This paper reveals details of our system that can be useful for further comparative study.

The SEC system represents highly modular structure, it is easily upgradeable, and can be connected to computer systems already existing at the university. The flexible architecture of the SEC allows adaptation both to the needs of the particular university and to changes in law and regulations. The SEC features wide functionality, including class assignment, cost analysis, marking, student portal, teaching and research management. The developers of the SEC – experienced academic teachers and software engineers – are at the same time the system users. This results in a well-thought-out system functions and structure, as well as the user-friendly interface. The implemented algorithms of data encryption and electronic signature meet the FIPS standards issued by NIST.

The aim of this paper is to describe functionality and architecture of the system developed at the Faculty for more than 10 years now. Its publication will help sharing experience by universities implementing such systems at their campuses. Since design, development and implementation of campus card systems are very complex tasks, we believe such information exchange is a necessary component of this enterprise. There is strong Europe-wide commitment to this approach, with emphasis on developing standards for information exchange, especially in view of the huge diversity in tradition of European academic institutions [21].

8.3 Functionality and Architecture of the System

The architecture of the system, with a superimposed diagram of teaching information flow is illustrated in Fig. 8.1. All the information collected in the Faculty is stored in a single Internet database. The Oracle 10g database management system [22] was selected in the TUL EECCE computer "Students' Electronic Card" (SEC) system design. The SEC system users save the information to and retrieve it from the database by means of terminals – computers equipped with a card reader and specialized program. In the case of the management of information related to teaching, as illustrated in Fig. 8.1, there are 3 types of terminals – dean office terminal (TD), teaching unit terminal (TJD) and teacher terminal (TN). Other terminals types are used to collect and/or process information related to research, to the cost analysis of the Faculty activities, cards management, etc., as will be shortly discussed later.

A student portal, which is in fact a web application, is also integrated with the system shown in Fig. 8.1. Two secure interfaces/channels that synchronize two separate data sets have been developed. A set of data which undergoes frequent modifications (e.g. lists of obligatory and elective courses, partial and final marks, dean's decisions, financial data) is fetched online directly from the central Oracle database. The data undergoing occasional modifications (e.g. student and teacher personal data, curricula, contact details, etc.) is buffered in a Postgres database and synchronized twice a day to the main Oracle system. The two data sets have been carefully defined to always ensure the integrity of the information presented in the student portal. Students can browse through their academic books of records (marks, enrollment to consecutive semesters, fees due and paid, etc.) after logging in to a password protected portal. An important function of the portal is in collecting students' answers to questionnaires on subjects, teachers, and organization of studies. Thanks to the portal integrated with the information management system, more than 80% of the students give feedback on the important aspects of the study to the Faculty.

In the future, once electronic student ID cards are issued to the students, the functionality of the student portal will be extended to e.g. electronic submission of study-related documents to the dean office. At present, cards are issued to academic and administration staff. The cards serve for visual identification of their users and for the storage of information about the range of user authorization privileges in the system.

The access to the card is protected by a secret personal identification number (PIN). Data written in to the database are encrypted [3] and signed digitally [4], [23]. The private keys needed for these operations are stored in the card memory.

A system terminal is a standard PC computer run under MS Windows® operating system, equipped with a card reader and dedicated software. An example of a card reader connected to a notebook computer through a serial USB port is shown in Fig. 8.2. The application window seen at the computer display is a dean office terminal (TD) window instance. Software for the terminals was written in Object Pascal and C++ languages using Delphi and C++ Builder development

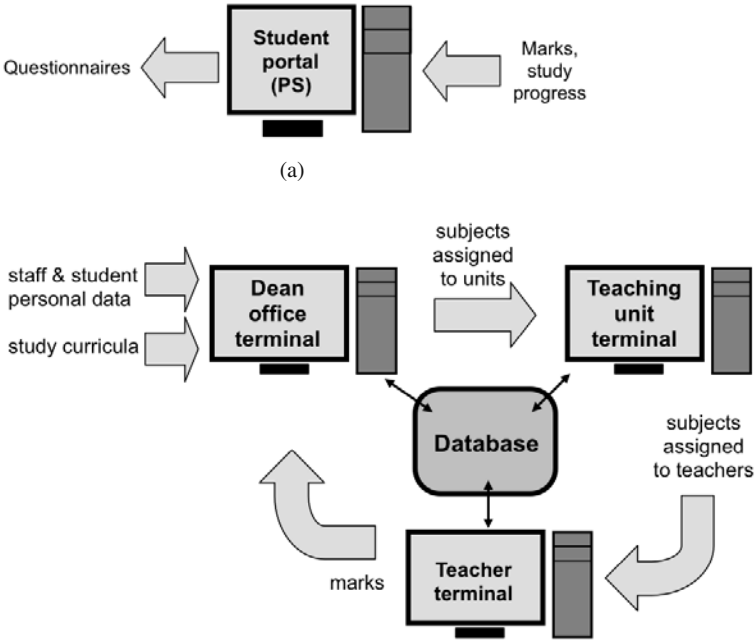

Fig. 8.1 SEC system architecture with an example of teaching-related flow of information

environment compilers [24], respectively. Operation of the student portal (PS) is encoded by PHP language [25] scripts run on an Apache server [26].

The electronic signature procedures are implemented by means of C++ and Object Pascal routines developed at the Faculty. The routines encode SHA-1 [27] and DSA [28] algorithms with the use of modular arithmetics [29]. Data encryption software modules use the TDEA (triple DES) algorithm [30].

Hybrid microprocessor cards Gemplus MPCOS-EMV R5 8000 (8kB) with Mifare contactless module (1kB) are currently used in the system [31], [32]. A special printer is applied to print the faculty and user-related information (Fig. 8.3) on both sides of the white plastic cards as they come from the vendor. A card issuing terminal (TWK) is used for card personalization. It comprises electronic encoding of

- data structure in the card memory,
- system information,
- user personal data,
- digital certificate of the card,

prior to graphic personalization.

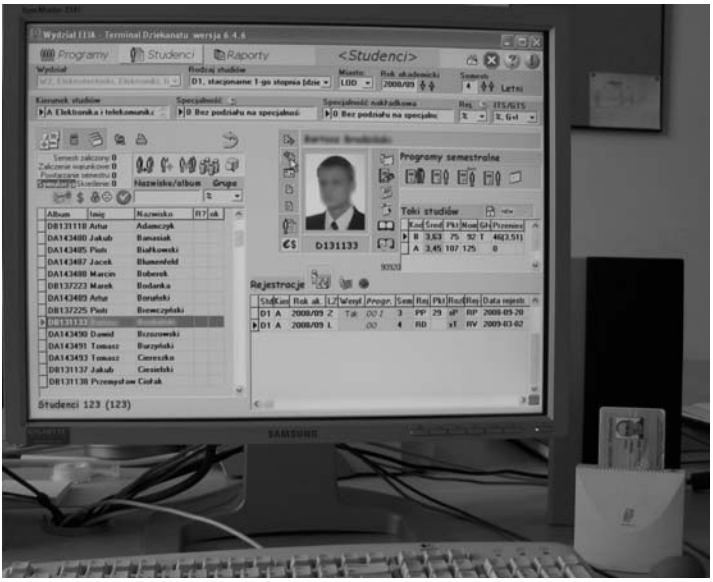

Fig. 8.2 Sample card reader (bottom right) connected to a PC terminal

Fig. 8.3 Example window of card issuing terminal (in Polish: terminal wydawania kart – TWK)

8.3.1 Cards and System Security

The data protection means built into the SEC information management system include [31]

- limited access to data (user authorization required, many user types defined with limited access rights),
- protection of the data content from unauthorized reading (encryption),
- measures of checking data integrity and authenticity (digital signature).

The basic tool of information protection is the microprocessor card. Its internal resources are utilized to store a) user identification data, b) cryptographic keys, c) electronic purse files, d) access control information.

The MPCOS-EMV cards allow implementation of various protection mechanisms of the access to stored data. The right to perform any operation on the contents of the files – reading, writing, modification – can be limited. Similar restrictions can be applied to creation of new elementary files (EF) and dedicated files, i.e. folders (DF). The card operating system includes two tests that have to be passed to get rights to perform required operations. One relies on verification of passwords, stored in SCEF – secret code elementary files. The other is based on verification of a cryptographic key by using a challenge-response method [3]. In the latter case, the card microprocessor generates a pseudorandom number and sends it to the computer application which has requested creation of a new folder. The application encrypts the received number using the 3DES algorithms based on asymmetric key shared with the card and sends the encrypted message to the card. The card compares the received message with the one generated locally by its microprocessor. If the messages are the same, the card performs the requested operation; otherwise, an error message is sent to the application.

The four layers of access protection to the card data files, implemented in the TUL EECCE information management system are illustrated in Fig. 8.4. There are six user categories, each attributed different rights to access the card data files and database records – teachers, dean office administrative staff, dean and dean deputies, teaching unit terminal operators, system administrators, and students. Independently of the user category, their cards carry the following information:

- user and card ID data,
- login names to system applications,
- unique private DSA key (for digital signature),
- unique private DES key (for encryption),
- user categories assigned to the card owner,
- electronic purse structure/data,
- login names and passwords for access control of campus facilities,
- card certificate.

On top of the above-described numerous security means, protocols and procedures built into the system, standard firewall technique of terminal communication with the database is implemented in it. In general,

- the level of information protection depends on the estimated cost of information loss or modification,

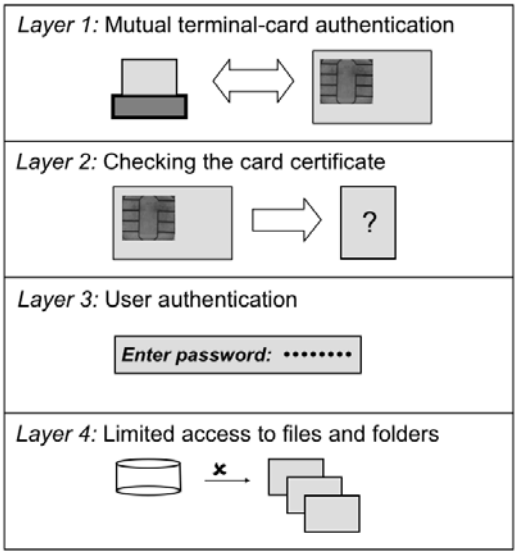

Fig. 8.4 Layers of access protection to data stored in the card memory

- the security mechanisms available with the Oracle DBMS are utilized as well (e.g. database users with different roles and privileges),
- terminal communication with cards is kept secret,
- digital signatures allow continuous verification of data authenticity and integrity.

The use of electronic cards in the Faculty information management system led to a data security level at which data eavesdropping or modification by unauthorized users was not possible, in practice. For more than 10 years of the system operation in the Faculty environment, no unauthorised data access was observed. The single-internet-database architecture with terminals located in the Faculty teaching units (institutes and departments) allows for on-line assignment of teaching load and continuous verification of the assignment consistency throughout the whole Faculty. For example, information about the actual number of students who have taken any subject class are easily available to a dean once teachers make proper choices from a list of student names generated by the system.

After the pilot studies and initial tests, the system was put into operation in 2002, to include year 1 students, studying under European credit transfer system (ECTS) [33]. Year by year, newly enrolled student records were added to the database. Since 2006, data of all the Faculty students have been stored and processed by the system. The benefits of using the system, in terms of speed of information collection and data consistency, as well as in availability of information which would otherwise not be possible to extract, are very clearly seen now. This positive evaluation made the Faculty decide to expand the system functionality to cover other strategic areas of its activity, beyond course assignment and subject marking.

8.3.2 Data Synchronization with the University Database

Since the very beginning of the system development, and especially after introduction of the ECTS study system at TUL in 2002, it was evident that there is (and always will be) the need for data exchange between the University central database and faculty local systems. The principal rule is that there should be only one source of information present in the whole University system. For example, the student ID number should be issued in the central system and copied to the local systems. On the other hand, marks given to students for their study performance are collected at faculties and sent to the central database. In the ECTS system, these marks are aggregates (e.g. weighted averages) of the marks student achieve for exams, laboratory classes or tutorials of a given subject, carried out by Faculty units. These partial marks are not relevant to the operation of the central system where aggregate marks are important only, e.g. taken into account to pick up the best students who deserve a scholarship award. On the contrary, teaching costs are generated in the units who run partial classes (lectures, labs, etc.), so it is very important to record all of them in the faculty local database, e.g. for operational cost analysis.

Well defined protocols and procedures for data exchange between local faculty systems and the central database are essential to smooth, efficient operation of the whole university information management system. The data should be exchanged in electronic form, manual copying of information from one system to the other is obviously obsolete and ridiculous.

8.3.3 System Expansion

The database in a faculty system is a reach source of information, essential to various aspects of the faculty activity. Often, there is a need to extract this information in a non-standard form, e.g. for statistical purposes. A reporting module was designed and written for this purpose (Fig. 8.5). Various information-selection criteria can be selected from a list corresponding to the information fields stored in the database. The module dynamically generates an SQL query which is sent to the base. The result (a report generated in response) is displayed on the computer screen and can be saved as an HTML or Excel file. This unique flexibility in report generation is a very useful feature, e.g. to satisfy the needs of statistical reports preparation requested by the Ministry of Education.

One of the essential factors of faculty management is identification of various sources of costs of running the teaching process. The cost components (staff wages, cost of overtime hours) should be evaluated and compared with the faculty income (yearly funds granted by ministry, student fees, etc.). One can then compute average costs of running specific courses or form of studies, e.g. evening courses, part time studies, for cost efficiency. Such data are the basis for strategic analyses, e.g. in the area of employment structure that matches the teaching quality criteria and gives maximum cost efficiency at the same time. To allow cost analysis in the SEC

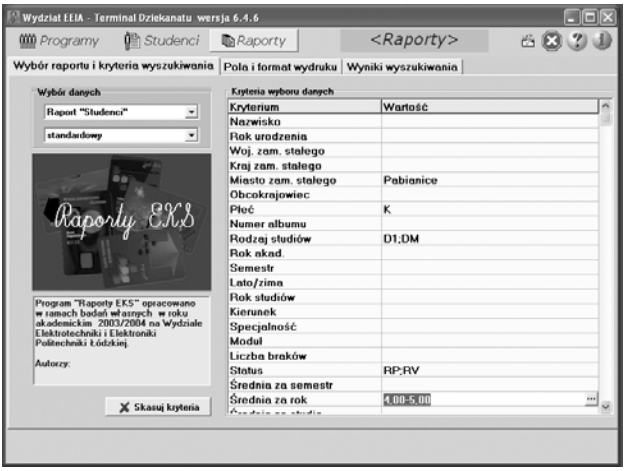

Fig. 8.5 Example window of flexible reporting program

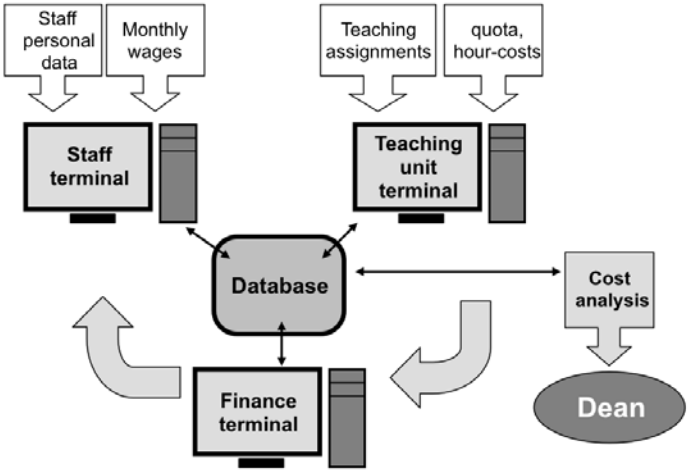

Fig. 8.6 Diagram of information flow for cost analysis

system, the system has been augmented with two terminals - staff terminal (TK with *Kadry* program) and finance terminal (TF with *Krezus* program), and the functionality of the teaching unit terminal (TJD, using the *Ekstazjusz* program) has been significantly enhanced. The terminals are used to collect and process information for the cost analysis task, as illustrated in Fig. 8.6.

The architecture of Fig. 8.1 was also adopted to apply the system for collecting and processing information related to the Faculty research activity. A new – research (TBN, Terminal Badań Naukowych in Polish) – terminal has been designed and implemented for the purpose, running the *Skryba* program. Depending on the user

privileges (read-out from the user card), the terminal changes its interface to allow writing in the information to the database (for officers representing the faculty units, see Fig. 8.7) or reading out the information only (for dean office workers).

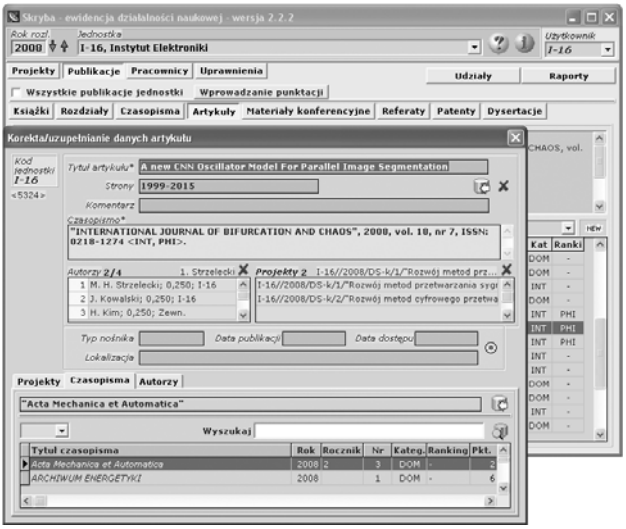

Fig. 8.7 Sample window for entering information on journal papers of TBN terminal

The TBN terminal allows collection of detailed information about research projects carried out in the institutes and departments, as well as on publications and various forms of research activity of all the staff members. Each form of research achievements is allocated a number of points which are counted to evaluate research activity in the faculty units. The points scored by a Faculty unit (institute or department) in a year are the basis for research funds allocation to it in the following year. Reports are generated, to make e.g. a list of publications for each unit, and the whole Faculty – for documentation required by the Ministry and Main Library, for staff assessment, planning, and other needs. A staff portal (based on an idea similar to the above described student portal) is under development now to provide individual staff members with an internet access to their research record data.

Another direction of the system expansion is in the area of card-based access control, e.g. to student laboratories. Both contact and contactless cards are considered in an in-progress Faculty project. Once electronic student ID cards are introduced within the University, they will be integrated with the existing system.

8.3.4 Design, Development, Update and Maintenance

The prototype system was designed and built in 1996-1998 by a 7-person team of experienced electronic engineers/computer programmers. Four of them had had

previous 2-4 years teaching work experience at foreign universities (USA, Australia, Japan, Great Britain) which was important in terms of introducing the credit point study system at the Faculty. Its design and deployment required solution of many complex engineering problems (functionality specification, database design, software development - user interfaces, card reader communication, algorithms for data encryption and digital signature) and, even more difficult, many logistic problems related to organization of the work of different groups of users (teachers, administration, students). For 10 years of the system operation, its functionality has been continuously evaluated (by a formal team of nominated users), updated and upgraded by a team of a few programmers. One of the recent projects in this area is "Electronic Education Connectivity Solution", aimed at providing support to Bologna processed-governed student mobility between European universities, funded in 2009-2011 from Seventh Framework European program resources [34].

8.4 Summary and Conclusion

The core element of the faculty information management SEC system at TUL EECCE Faculty is a microprocessor card for the storage of user's personal information and private keys for data encryption and digital signature. The card provides high security level necessary for safe communication with a single database. Single source of information allows fast access to it and eliminates the very high costs of verification and processing of the data collected in a paper form. The introduction of the system contributed much to the quality of the Faculty work. Some vital decision-support analyses were not possible at all before, e.g. detailed cost evaluation of average teaching hour, say, at evening courses in satellite TUL campuses. Implementation of a campus card system is a complex task, not only from the technical, but mainly organizational point of view. We found it essential that users of the university systems were involved in the process of setting system specification, testing and evaluation of new hardware and software modules. It seams impossible that an external company, with no work experience in teaching and research organization can really succeed in development of a system that seamlessly integrates with faculty complex activities and helps in performing the faculty tasks rather than being a problem by itself.

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